

# Question Paper

Exam Date & Time: 18-Nov-2025 (02:00 PM - 05:00 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU  
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Soft Computing Techniques [ELE\_4410]

Marks: 50

Duration: 180 mins.

### PART-A

Answer all the questions.

- 1) Describe any two activation functions used in training artificial neural networks with necessary mathematical expression and its graphical representation? (5)  
A) 

Course Outcome: "CO1" / Cognitive Level: "Understand"

  
B) Illustrate the winner take-all learning method used in learning of neural networks with computational steps. (5)  

Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Analyse the error backpropagation algorithm used in training the multilayer feed-forward neural network with necessary computational steps. (5)  
A) 

Course Outcome: "CO2" / Cognitive Level: "Application"

  
B) Explain the RBF kernel for modelling multivariate data with expression. Explain the interpretation of covariance matrix used in it. (5)  

Course Outcome: "CO2" / Cognitive Level: "Understand"
- 3) Build the fuzzy Sugino inference model for decision making with computational steps. (5)  
A) 

Course Outcome: "CO3" / Cognitive Level: "Application"

  
B) Describe and construct convex and non-convex fuzzy sets used in fuzzy inference with an example each along with its membership function. (5)  

Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Analyse crossover and mutation operators used in genetic algorithms as operators. Explain the analogy of these operators in natural evolution process of human. (5)  
A) 

Course Outcome: "CO4" / Cognitive Level: "Analysis"

  
B) Explain how a minimization problem in optimization can be carried out using genetic algorithm by maximizing the fitness function. Explain in detail. (5)  

Course Outcome: "CO4" / Cognitive Level: "Application"
- 5) With a neat architecture, explain the adaptive neuro-fuzzy inference system (ANFIS)? (5)

A) Course Outcome: "CO5" / Cognitive Level: "Application"

B) Differentiate between fuzzy neural network and ANFIS, used as decision support systems? (5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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| Event Name | Subject |
|------------|---------|
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058-END SEMSTER-ODD-2025 ELE\_4410:Soft Computing Techniques

| #            | Course Outcome | Description | Marks     |
|--------------|----------------|-------------|-----------|
| 1            | 1              | CO 1        | 10        |
| 2            | 2              | CO 2        | 10        |
| 3            | 3              | CO 3        | 10        |
| 4            | 4              | CO 4        | 10        |
| 5            | 5              | CO 5        | 10        |
| <b>Total</b> |                |             | <b>50</b> |

\* **NA** - CO Not Selected

| Event Name | Subject |
|------------|---------|
|------------|---------|

058-END SEMSTER-ODD-2025 ELE\_4410:Soft Computing Techniques

| #     | Level       | No of Questions | Marks (%) |
|-------|-------------|-----------------|-----------|
| 1     | Remember    | 0               | 0(0%)     |
| 2     | Understand  | 2               | 10(20%)   |
| 3     | Application | 5               | 25(50%)   |
| 4     | Analysis    | 3               | 15(30%)   |
| 5     | Create      | 0               | 0(0%)     |
| 6     | Evaluate    | 0               | 0(0%)     |
| Total |             | 10              | 50 (100%) |